

Jenkins Shared Libraries

Building Reusable Enterprise Pipelines

Topics

- Shared Libraries
- vars
- src
- resources
- Global Libraries
- Folder Libraries
- Best Practices



Why Shared Libraries?

Problem

Pipeline A 500 lines Pipeline B 450 lines Pipeline C 600 lines

Lots of duplicate code.

Solution

Move common logic into Shared Libraries.

Benefits

✓ Reusable

✓ Standardized

✓ Easy Maintenance

✓ Version Controlled

✓ Enterprise Ready

Library Architecture



One library

Many projects

Shared Library Folder Structure

```
shared-library/
```

```
├─ vars/
```

```
├─ src/
```

```
├─ resources/
```

```
└─ README.md
```

Explain every folder.

vars Directory

Purpose	Example	Usage
Global Pipeline Functions	buildApp.groovy	buildApp()

Example vars Function

```
def call(){  
  echo "Building Application"  
}
```

Pipeline

```
buildApp()
```

src Directory

Purpose

Groovy Classes

Example

```
src/  
  com/  
    company/  
      Build.groovy
```


Example Groovy Class

```
class Build{  
    def compile(){  
    }  
}
```

Explain Object Oriented approach.

resources Directory

Store

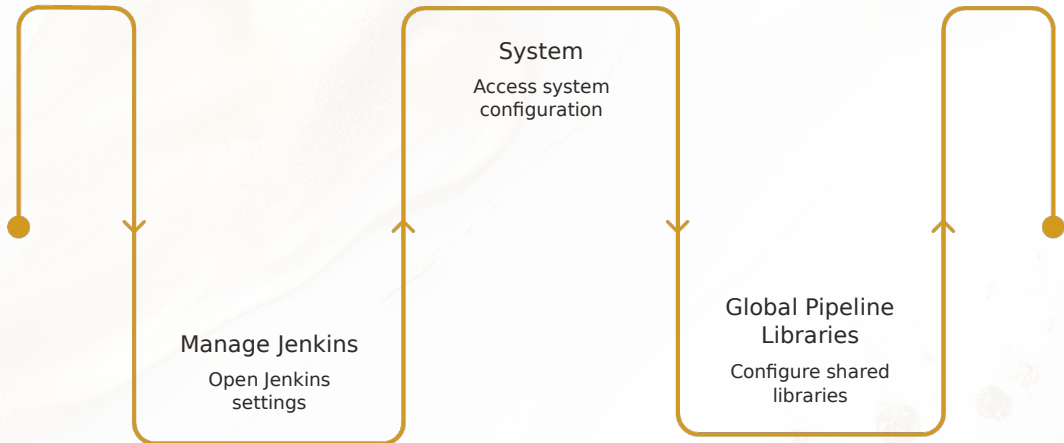
- YAML
- JSON
- Templates
- Scripts

Use

```
libraryResource()
```

Global Shared Library

Navigate



Configure

- Name
- Default Version
- Retrieval Method



Folder Shared Library

Explain

Libraries available only inside one folder.

Enterprise use case.

Loading Libraries

```
@Library('shared-lib') _
```

Dynamic

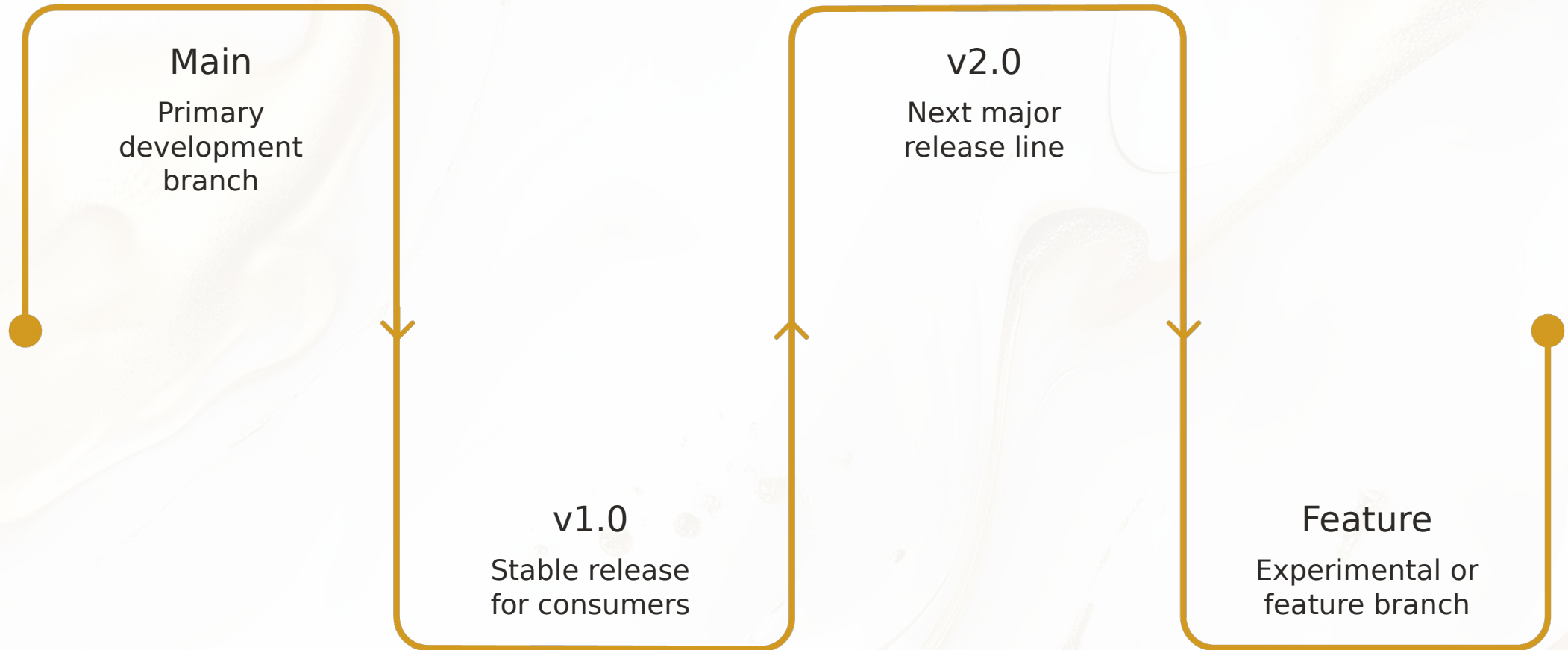
```
library 'shared-lib'
```


Calling Functions

```
buildApp()  
testApp()  
deployApp()
```

Simple and clean.

Library Versioning



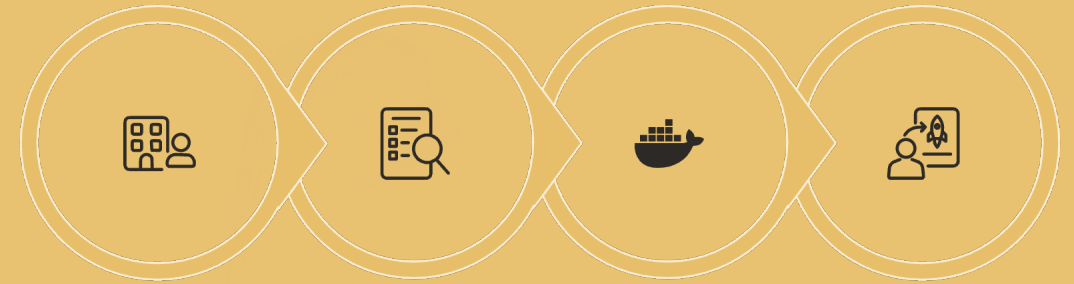
Pin Pipelines to specific versions.

Enterprise Example

Instead of

500-line Jenkinsfile

Use



build()

test()

docker()

deploy()

Each function lives inside Shared Library.

Best Practices

Small functions

Documentation

Semantic versioning

Unit testing

Separate repositories

Common Mistakes

Huge libraries

Circular dependencies

Hardcoded paths

No version control

Enterprise Pipeline

```
@Library('company-lib')  
pipeline{  
    Build()  
    Test()  
    Docker()  
    Deploy()  
}
```

Very small Jenkinsfile.